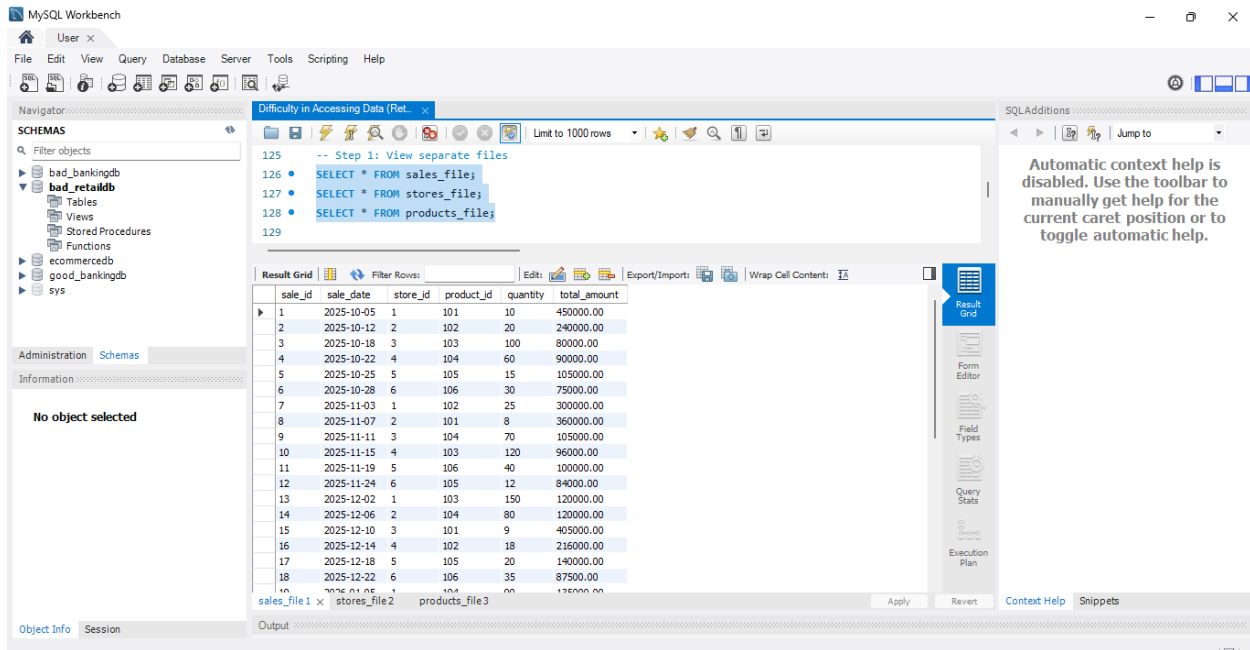


Name: Chit Hmue Than Thar

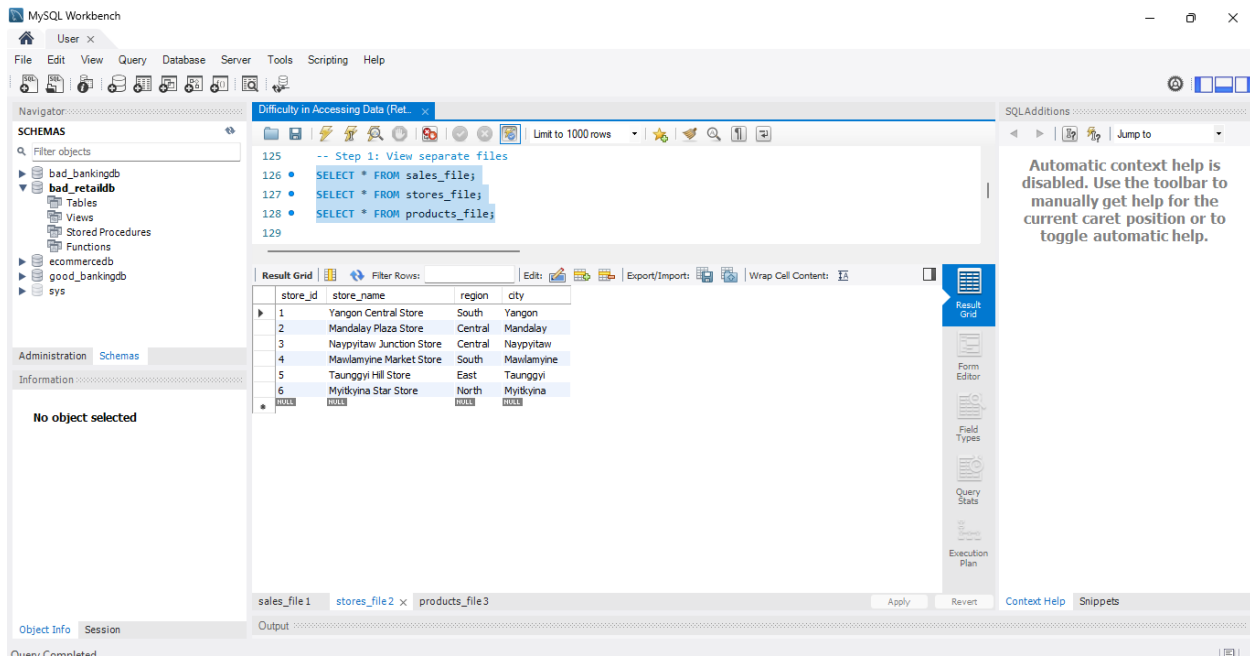
Enrollment ID: DE-2026-0186



MySQL Workbench interface showing a query result grid. The query is: `-- Step 1: View separate files`
`SELECT * FROM sales_file;`
`SELECT * FROM stores_file;`
`SELECT * FROM products_file;`

The result grid displays the following data:

sale_id	sale_date	store_id	product_id	quantity	total_amount
1	2025-10-05	1	101	10	450000.00
2	2025-10-12	2	102	20	240000.00
3	2025-10-18	3	103	100	80000.00
4	2025-10-22	4	104	60	90000.00
5	2025-10-25	5	105	15	105000.00
6	2025-10-28	6	106	30	75000.00
7	2025-11-03	1	102	25	300000.00
8	2025-11-07	2	101	8	360000.00
9	2025-11-11	3	104	70	105000.00
10	2025-11-15	4	103	120	96000.00
11	2025-11-19	5	106	40	100000.00
12	2025-11-24	6	105	12	84000.00
13	2025-12-02	1	103	150	120000.00
14	2025-12-06	2	104	80	120000.00
15	2025-12-10	3	101	9	405000.00
16	2025-12-14	4	102	18	216000.00
17	2025-12-18	5	105	20	140000.00
18	2025-12-22	6	106	35	87500.00



MySQL Workbench interface showing a query result grid. The query is: `-- Step 1: View separate files`
`SELECT * FROM sales_file;`
`SELECT * FROM stores_file;`
`SELECT * FROM products_file;`

The result grid displays the following data:

store_id	store_name	region	city
1	Yangon Central Store	South	Yangon
2	Mandalay Plaza Store	Central	Mandalay
3	Naypyitaw Junction Store	Central	Naypyitaw
4	Mawlamyine Market Store	South	Mawlamyine
5	Taunggyi Hill Store	East	Taunggyi
6	Myitkyina Star Store	North	Myitkyina

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret...

Limit to 1000 rows

-- Step 1: View separate files

```
125 -- Step 1: View separate files
126 SELECT * FROM sales_file;
127 SELECT * FROM stores_file;
128 SELECT * FROM products_file;
129
```

Result Grid

product_id	product_name	category	unit_price
101	Rice Bag	Grocery	45000.00
102	Cooking Oil	Grocery	12000.00
103	Instant Noodles	Grocery	800.00
104	Soft Drink	Beverage	1500.00
105	Shampoo	Personal Care	7000.00
106	Soap	Personal Care	2500.00

sales_file 1 stores_file 2 products_file 3 x

Apply Revert Context Help Snippets

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret...

Limit to 1000 rows

```
133 s.store_id,
134 st.store_name,
135 st.region,
136 s.total_amount
137 FROM sales_file s
138 JOIN stores_file st ON s.store_id = st.store_id
139 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
140 ORDER BY st.region, s.sale_date;
```

Result Grid

sale_date	store_id	store_name	region	total_amount
2025-10-12	2	Mandalay Plaza Store	Central	240000.00
2025-10-18	3	Naypyitaw Junction Store	Central	80000.00
2025-11-07	2	Mandalay Plaza Store	Central	360000.00
2025-11-11	3	Naypyitaw Junction Store	Central	105000.00
2025-12-06	2	Mandalay Plaza Store	Central	120000.00
2025-12-10	3	Naypyitaw Junction Store	Central	405000.00
2026-01-08	2	Mandalay Plaza Store	Central	112000.00
2026-01-12	3	Naypyitaw Junction Store	Central	264000.00
2026-02-07	2	Mandalay Plaza Store	Central	125000.00

Result 4 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
83	00:55:13	SELECT * FROM products_file LIMIT 0.1000	6 row(s) returned	0.000 sec / 0.000 sec
84	01:00:36	SELECT s.sale_date, s.store_id, st.store_name, st.region, s.total_amount	36 row(s) returned	0.016 sec / 0.000 sec

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information: No object selected

Difficulty in Accessing Data (Ret...)

```

190 -- Sales report for last 6 months by region
191 SELECT
192     st.region,
193     SUM(s.total_amount) AS total_sales
194 FROM sales s
195 JOIN stores st ON s.store_id = st.store_id
196 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
197 GROUP BY st.region
198 ORDER BY total_sales DESC;
199
200 -- Monthly sales report by region
201 SELECT

```

Result Grid

region	total_sales
South	2268500.00
Central	2165500.00
East	1229500.00
North	1093500.00

Result 5 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
95	01:09:37	INSERT INTO good_retaildb sales (sale_date, store_id, product_id, quantity, total_amo...	36 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.016 sec
96	01:10:13	SELECT st.region, SUM(s.total_amount) AS total_sales FROM sales s JOIN store...	4 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Object Created

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator: Schemas

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information: No object selected

Difficulty in Accessing Data (Ret...)

```

200 -- Monthly sales report by region
201 SELECT
202     DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month,
203     st.region,
204     SUM(s.total_amount) AS total_sales
205 FROM sales s
206 JOIN stores st ON s.store_id = st.store_id
207 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
208 GROUP BY DATE_FORMAT(s.sale_date, '%Y-%m'), st.region
209 ORDER BY sales_month, st.region;

```

Result Grid

sales_month	region	total_sales
2025-10	Central	320000.00
2025-10	East	105000.00
2025-10	North	75000.00
2025-10	South	540000.00
2025-11	Central	465000.00
2025-11	East	100000.00
2025-11	North	84000.00
2025-11	South	100000.00

Result 6 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
96	01:10:13	SELECT st.region, SUM(s.total_amount) AS total_sales FROM sales s JOIN store...	4 row(s) returned	0.015 sec / 0.000 sec
97	01:12:27	SELECT DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month, st.region, S...	24 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Object Created

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Object Info Session

Query Completed

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```
200 -- Monthly sales report by region
201 • SELECT
202     DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month,
203     st.region,
204     SUM(s.total_amount) AS total_sales
205 FROM sales s
206 JOIN stores st ON s.store_id = st.store_id
207 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
208 GROUP BY DATE_FORMAT(s.sale_date, '%Y-%m'), st.region
209 ORDER BY sales_month, st.region;
```

Result Grid

sales_month	region	total_sales
2026-02	East	495000.00
2026-02	North	192000.00
2026-02	South	254000.00
2026-03	Central	242000.00
2026-03	East	252000.00
2026-03	North	585000.00
2026-03	South	292500.00

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
96	01:10:13	SELECT st.region, SUM(s.total_amount) AS total_sales FROM sales s JOIN store...	4 row(s) returned	0.015 sec / 0.000 sec
97	01:12:27	SELECT DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month, st.region, S...	24 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Object Info Session

Query Completed

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```
212 • SELECT
213     st.region,
214     p.category,
215     SUM(s.total_amount) AS total_sales
216 FROM sales s
217 JOIN stores st ON s.store_id = st.store_id
218 JOIN products p ON s.product_id = p.product_id
219 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31';
```

Result Grid

region	category	total_sales
Central	Grocery	1605000.00
Central	Beverage	337500.00
Central	Personal Care	223000.00
East	Grocery	747000.00
East	Personal Care	482500.00
North	Grocery	777000.00
North	Personal Care	316500.00
South	Grocery	1625000.00
South	Beverage	367500.00
South	Personal Care	276000.00

Result 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	01:12:27	SELECT DATE_FORMAT(s.sale_date, '%Y-%m') AS sales_month, st.region, S...	24 row(s) returned	0.000 sec / 0.000 sec
98	01:35:18	SELECT st.region, p.category, SUM(s.total_amount) AS total_sales FROM sal...	10 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
- Tables
- Views
- Stored Procedures
- Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```

237 s.quantity,
238 s.total_amount
239 FROM sales s
240 JOIN stores st ON s.store_id = st.store_id
241 JOIN products p ON s.product_id = p.product_id;
242

```

Result Grid

sale_id	sale_date	store_id	store_name	region	city	product_id	product_name	category	quantity
1	2025-10-05	1	Yangon Central Store	South	Yangon	101	Rice Bag	Grocery	10
7	2025-11-03	1	Yangon Central Store	South	Yangon	102	Cooking Oil	Grocery	25
13	2025-12-02	1	Yangon Central Store	South	Yangon	103	Instant Noodles	Grocery	150
19	2026-01-05	1	Yangon Central Store	South	Yangon	104	Soft Drink	Beverage	90
25	2026-02-03	1	Yangon Central Store	South	Yangon	105	Shampoo	Personal Care	18
31	2026-03-04	1	Yangon Central Store	South	Yangon	106	Soap	Personal Care	60
2	2025-10-12	2	Mandalay Plaza Store	Central	Mandalay	102	Cooking Oil	Grocery	20
8	2025-11-07	2	Mandalay Plaza Store	Central	Mandalay	101	Rice Bag	Grocery	8
14	2025-12-06	2	Mandalay Plaza Store	Central	Mandalay	104	Soft Drink	Beverage	80
20	2026-01-08	2	Mandalay Plaza Store	Central	Mandalay	103	Instant Noodles	Grocery	140
26	2026-02-07	2	Mandalay Plaza Store	Central	Mandalay	106	Soap	Personal Care	50

vw_sales_reporting8 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
100	01:38:33	CREATE VIEW vw_sales_reporting AS SELECT s.sale_id, s.sale_date, st.store...	Error Code: 1050. Table 'vw_sales_reporting' already exists	0.000 sec
101	01:38:39	SELECT * FROM vw_sales_reporting LIMIT 0, 1000	36 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
- Tables
- Views
- Stored Procedures
- Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```

247 -- Raw sales file does not directly show region
248 USE bad_retaildb;
249 SELECT
250 sale_id,
251 sale_date,
252 store_id,
253 total_amount
254 FROM sales_file
255 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31';

```

Result Grid

sale_id	sale_date	store_id	total_amount
1	2025-10-05	1	450000.00
2	2025-10-12	2	240000.00
3	2025-10-18	3	80000.00
4	2025-10-22	4	90000.00
5	2025-10-25	5	105000.00
6	2025-10-28	6	75000.00
7	2025-11-03	1	300000.00
8	2025-11-07	2	360000.00
9	2025-11-11	3	100000.00

sales_file9 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
102	01:39:30	USE bad_retaildb	0 row(s) affected	0.000 sec
103	01:39:51	SELECT sale_id, sale_date, store_id, total_amount FROM sales_file WHERE...	36 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator: Difficulty in Accessing Data (Ret...)

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information: No object selected

```
-- Test Case 3: DBMS quickly answers manager's question
268 USE good_retaildb;
269
270 SELECT
271     st.region,
272     SUM(s.total_amount) AS total_sales
273 FROM sales s
274 JOIN stores st ON s.store_id = st.store_id
275 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31'
276 GROUP BY st.region
277 ORDER BY total_sales DESC;
278
```

Result Grid

	region	total_sales
▶	South	2268500.00
	Central	2165500.00
	East	1229500.00
	North	1093500.00

Result 12 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
106	01:42:05	USE good_retaildb	0 row(s) affected	0.000 sec
107	01:42:21	SELECT st.region, SUM(s.total_amount) AS total_sales FROM sales s JOIN store...	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator: Difficulty in Accessing Data (Ret...)

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information: No object selected

```
259 SELECT
260     s.sale_id,
261     s.sale_date,
262     st.region,
263     s.total_amount
264 FROM sales_file s
265 JOIN stores_file st ON s.store_id = st.store_id
266 WHERE s.sale_date BETWEEN '2025-10-01' AND '2026-03-31';
267
```

Result Grid

	sale_id	sale_date	region	total_amount
▶	1	2025-10-05	South	450000.00
	2	2025-10-12	Central	240000.00
	3	2025-10-18	Central	80000.00
	4	2025-10-22	South	90000.00
	5	2025-10-25	East	105000.00
	6	2025-10-28	North	75000.00
	7	2025-11-03	South	300000.00
	8	2025-11-07	Central	360000.00
	9	2025-11-11	Central	105000.00

Result 11 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
104	01:41:01	SELECT s.sale_id, s.sale_date, st.region, s.total_amount FROM sales_file s ...	36 row(s) returned	0.000 sec / 0.000 sec
105	01:41:11	SELECT s.sale_id, s.sale_date, st.region, s.total_amount FROM sales_file s ...	36 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
- Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```
280
281 • SELECT
282     region,
283     COUNT(*) AS transaction_count,
284     SUM(total_amount) AS total_sales
285 FROM vw_sales_reporting
286 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
287 GROUP BY region
288 ORDER BY total_sales DESC;
289
```

Result Grid

region	transaction_count	total_sales
South	12	2268500.00
Central	12	2165500.00
East	6	1229500.00
North	6	1093500.00

Result 14 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
108	01:43:53	SELECT region, COUNT(*) AS transaction_count, SUM(total_amount) AS total...	4 row(s) returned	0.000 sec / 0.000 sec
109	01:44:16	SELECT region, COUNT(*) AS transaction_count, SUM(total_amount) AS total...	4 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

MySQL Workbench

User x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- bad_bankingdb
- bad_retaildb
 - Tables
 - Views
 - Stored Procedures
- Functions
- ecommercedb
- good_bankingdb
- sys

Administration Schemas

Information

No object selected

Difficulty in Accessing Data (Ret. x)

Limit to 1000 rows

```
290 -- Test Case 5: Monthly trend by region
291 • SELECT
292     DATE_FORMAT(sale_date, '%Y-%m') AS sales_month,
293     region,
294     SUM(total_amount) AS total_sales
295 FROM vw_sales_reporting
296 WHERE sale_date BETWEEN '2025-10-01' AND '2026-03-31'
297 GROUP BY DATE_FORMAT(sale_date, '%Y-%m'), region
298 ORDER BY sales_month, region;
```

Result Grid

sales_month	region	total_sales
2025-10	Central	320000.00
2025-10	East	105000.00
2025-10	North	75000.00
2025-10	South	540000.00
2025-11	Central	465000.00
2025-11	East	100000.00
2025-11	North	84000.00
2025-11	South	396000.00

Result 15 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
109	01:44:16	SELECT region, COUNT(*) AS transaction_count, SUM(total_amount) AS total...	4 row(s) returned	0.015 sec / 0.000 sec
110	01:45:26	SELECT DATE_FORMAT(sale_date, '%Y-%m') AS sales_month, region, SUM...	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session